

ANIKET KUNDU

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EDUCATION

Vellore Institute of Technology, Chennai

(07/2023 – 05/2027)

Bachelor of Technology in Computer Science Engineering

CGPA : 8.64/10, Highest SGPA: 9.21/10

WORK EXPERIENCE/INTERNSHIPS

SpectoV | AI/Backend Engineer (Initial Team)

(09/2024 – 02/2025)

- Led a team building an automated data pipeline using Google Cloud Platform API services, optimizing background worker processes for logistics records.
- Developed conversational AI agents for B2C interview preparation and implemented a multi-document loader RAG system using Hugging Face open-source LLMs and the GROQ API for B2B solutions.

Chakaralaya Analytics | Software Engineer Intern

(06/2024 – 08/2024)

- Worked on conjoint analysis under the guidance of the cofounder and CEO *Dr.Sudarsanam S.K.* from IIT Madras
- Used PyTorch to develop a highly optimised GPU-accelerated Multi-Criteria Decision Making (MCDM) library (to replace the slower PyDecision Library) along with JSON support for the Supplier Selection backend.

ACHIEVEMENTS and CERTIFICATIONS

- International Collegiate Programming Contest (ICPC) Asia-West Regionals 2025-26 Qualified:**
Prelims Institute rank 2, Regionals Rank: 160 out of 350 teams icpc.global/icpcid/0FI3JZCNH2Q0
- Codeforces (Expert):** Highest Rating 1600+, Solved 400+ problems codeforces.com/profile/Anicetus_7
- LeetCode (Knight):** Highest Rating 2059+(Top 2.5% worldwide), 750+ problems(120+ Hard)
leetcode.com/u/Anicetus_7
- SOF NCO Cyber Olympiad 2019:** International Rank 18, Zonal Rank 1
- MongoDB Database Administrator Certification

PROJECTS

Z-BFT | Modern C++, Distributed Systems, Cryptography, Asio | (on-going semester project)

A Zero-trust, BFT Consensus-backed, Encrypted key-value store architecture

- Architected a **Zero-Trust Identity Gateway** acting as a high-speed ingress firewall, utilizing probabilistic **Count-Min Sketches** for $O(1)$ memory rate-limiting to neutralize DDoS attacks and stateful memory exhaustion.
- Engineered an **Optimistic Byzantine Fault Tolerant (BFT) Consensus** Cluster to maintain state replication, gracefully degrading from ultra-fast symmetric **HMACs** to heavy asymmetric **Ed25519 signatures** upon anomaly detection.
- Built an ultra-low latency, in-memory **Distributed Key-Value Store** leveraging a custom **Arena Allocator** to bypass OS heap fragmentation and ensure predictable, high-throughput data retrieval.
- Integrated **TTL-Adaptive AES-NI** Hardware Encryption via x86 compiler intrinsics, **dynamically scaling cryptographic rounds** based on data lifespan to secure data-at-rest with near-zero CPU cycle penalties.
- Validated architectural resilience and throughput using deterministic testing harnesses, applying **industry-standard benchmarking pipelines** using tools like **Valgrind, gprof, or perf** to empirically map latency-versus-security trade-offs under simulated fault injection.

DistW | Modern C++, Distributed Systems, Concurrency, uWebSockets

A collaborative IDE featuring hierarchical locking for granular distributed workflow

- Architected a real-time collaborative IDE workspace in **C++20**, utilizing **uWebSockets** for asynchronous I/O multiplexing to handle high-frequency keystroke broadcasts and low-latency state synchronization across remote teams.
- Engineered a Hierarchical Distributed Lock Manager (HDLM) using a **Trie-based prefix tree** to solve collaborative edit collisions. This enforces pessimistic concurrency (granular file/folder locks) to guarantee atomic data integrity without the memory bloat of CRDTs (Conflict-free Replication Data Types).

- Developed a Delta Engine powered by a **vectorized line buffer** to achieve amortized **O(1) line-level insertions**, enabling bandwidth-efficient text synchronization and Google Docs-style real-time multiplayer cursor tracking.
- Built a pre-warmed **Docker execution sandbox** to safely compile and run untrusted user code. Achieved **sub-300ms cold starts** while using an Admin Dashboard to enforce strict resource caps (CPU/RAM/PIDs) to prevent malicious host exhaustion.
- Implemented a multi-threaded **Reaper Daemon** to prevent memory leaks and zombie processes by auto-purging disconnected user sessions and dormant containers, enabling the server to scale efficiently to **1,000+ concurrent connections**.

AeroHedge  |

Modern C++, Lock-Free Concurrency, Distributed I/O

A Zero-Allocation Option Hedging Engine with Algorithmic Simulation

- Architected a deterministic, high-frequency options trading engine in **Modern C++20**, enforcing strict **zero-allocation** (heap-bypass) policies and branchless polynomial mathematics on the critical path to achieve sub-microsecond tick-to-trade execution.
- Designed a lock-free concurrency model using **SPSC ring buffers** with strict **atomic memory ordering** (acquire/release), bypassing OS kernel scheduler context switches. Eradicated hardware false sharing via explicit **64-byte L1 cache-line alignment** across explicitly pinned CPU cores.
- Engineered a high-throughput distributed network layer utilizing **Zero-Copy UDP multicast** deserialization directly into optimally packed memory structs. Implemented a **non-blocking TCP IPv4 gateway** utilizing user-space spin-polling to aggressively manage exchange backpressure without yielding the CPU.
- Built an out-of-band UDP **fire-and-forget pipeline** bridged to an asynchronous Python/WebSocket server, streaming nanosecond-precision **RDTS (Read Time-Stamp Counter)** hardware metrics to a UI without degrading core determinism.

COURSEWORK and SKILLS

Languages: Modern C++ (17+), Python, Oracle SQL, C, Java, Bash shell scripting

Tools and Libraries: NumPy, CMAKE, C++ STL and Modern Concurrency and Networking Libraries

Technologies/Frameworks: Git, GitHub, Django REST, Amazon Web Services

Coursework: High Performance Computing, Data Structures & Algorithms, Design & Analysis of Algorithms, Computer Architecture, Operating Systems, Complex Variables & Linear Algebra, Probability & Statistics, Natural Language Processing, AWS Cloud Architecture